

Engine Only Operates in Droop Mode

Symptoms

- The engine will **only** operate in droop mode.

How To Use This Tree

This symptom tree can be used to troubleshoot engine droop symptoms. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

None.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the customer interface box (C.I.B.) wiring.	
	STEP 1A. Check the droop adjust potentiometer SUPPLY and SIGNAL wires for a short circuit.	Greater than 100k ohms?
STEP 2.	Check the engine harness to the C.I.B.	
	STEP 2A. Check the droop adjust potentiometer SUPPLY and SIGNAL wires.	Less than 10 ohms?

STEP 1. Check the customer interface box (C.I.B.) wiring.

STEP 1A. Check the droop adjust potentiometer SUPPLY and SIGNAL wires for a short circuit.

Conditions :

- Open the C.I.B.
- Disconnect the C.I.B. to engine harness cable connector C1 from the C.I.B.

Action	Specification/Repair	Next Step
Check the droop adjust potentiometer SUPPLY and SIGNAL wires. • Disconnect the droop adjust 5 volt SUPPLY wire and the droop adjust potentiometer SIGNAL wire from the X1 connector. • Place one test lead on the droop adjust potentiometer 5 volt SUPPLY pin in connector C1. • Place the other test lead on the droop adjust potentiometer SIGNAL pin in connector C1.	Greater than 100k ohms? YES Repair : Replace the faulty wire(s). Refer to Procedure 015-023 in Section 15. (/qs3/pubsys2/xml/en/procedures/300/300-015-023.html)	Repair complete
	Greater than 100k ohms? NO	2A

STEP 2. Check the engine harness to the C.I.B.**STEP 2A. Check the droop adjust potentiometer SUPPLY and SIGNAL wires.****Conditions :**

- Disconnect the C.I.B. to the engine harness cable connector C1 from the C.I.B.
- Disconnect the C.I.B. to the engine harness cable connector C4 from the engine harness.

Action	Specification/Repair	Next Step
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Check the droop adjust potentiometer SUPPLY and SIGNAL wires. - Place one test lead in the droop adjust potentiometer SUPPLY pin of the C1 connector. - Place the other test lead in the droop adjust potentiometer SIGNAL pin of the C1 connector.	Less than 10 ohms? YES Repair : Replace the cable.	Repair complete
	Less than 10 ohms? NO Repair : Use the following for potentiometer repair instructions. Reference the Marine Auxiliary QSB7-DM CM850 Fault Code Troubleshooting Manual, Bulletin 4325972, Section TF; the ISM and QSM11 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666266, Section TF; X15 CM2350 X125M Fault Code Troubleshooting Manual, Bulletin 5504346, Section TF; or the equipment manufacturer service information.	Repair complete